Continuous Inventory: Technology Strategy

Status: Draft for review **Date:** 27 September 2026 **Scope:** Current state, product vision, strategic capabilities, operating principles, and success criteria **Next stage:** Implementation architecture after strategy approval

Executive summary

The current platform provides substantial inventory, manufacturing, fulfillment, and financial transaction capabilities. The next opportunity is to make those capabilities work continuously on the customer's behalf: anticipate operational needs, prepare coordinated actions, secure appropriate authorization, carry work through execution, and verify the result.

Perpetual inventory keeps recorded inventory current as transactions occur. Our proposed continuous inventory system extends this into continuous operational management. It connects demand, available inventory, purchasing, manufacturing, warehouse execution, fulfillment, and cost into an ongoing decision-and-execution cycle.

The primary customer experience is **done for you, with approval**. A customer receives a complete, evidence-backed proposal that is ready to execute. Once approved, the system owns coordination and follow-through. Over time, customers can explicitly delegate bounded classes of actions as performance establishes trust.

The system must learn how each customer operates, at business, item, location, and seasonal levels. It should support bulk purchasing and production, purchase/build-to-order, and combinations of these practices within the same business. Customers can teach the agent their process from day one and correct its interpretation as conditions change. Customer-specific learning and explicit steering are foundational to every experience, including customers with no usable transaction history.

This strategy builds on Item Master, Inventory Operations Management (IOM), One Transaction Service (1TS), and the existing reporting, search, and audit capabilities. It preserves their authoritative responsibilities. The new capability coordinates decisions and outcomes across them; it does not establish a competing inventory ledger or financial system.

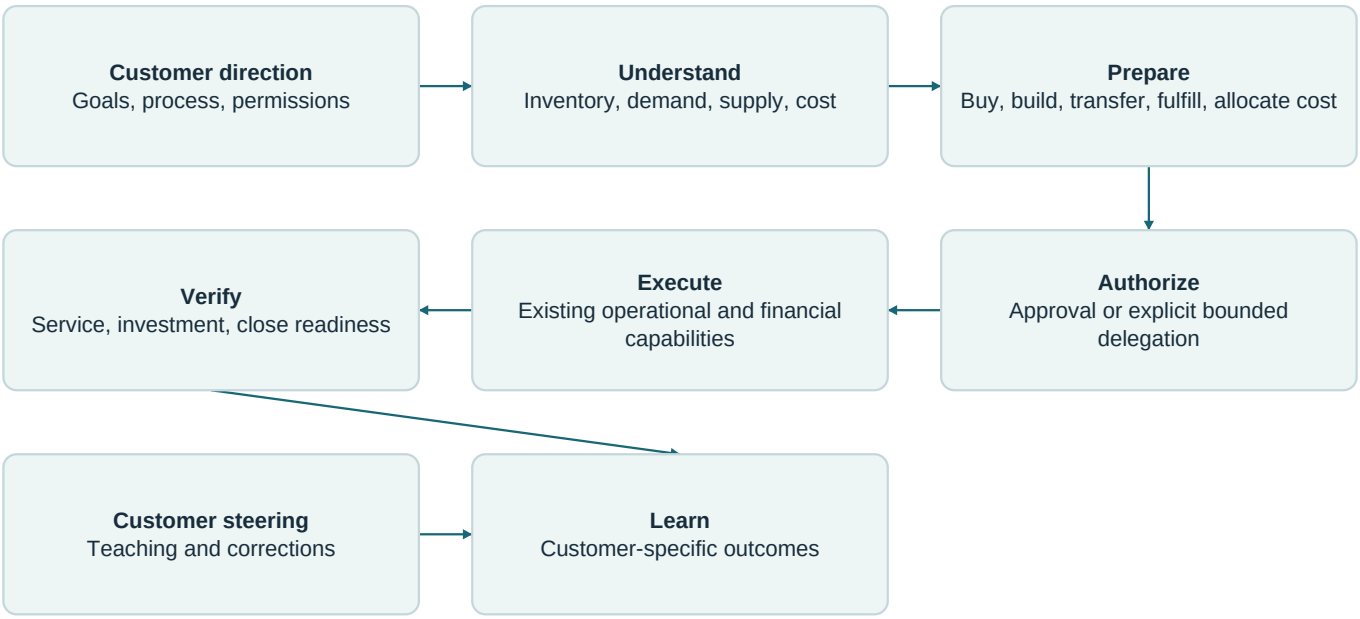
The strategic focus is eight connected experiences: automatic replenishment, automatic build planning, sales-order routing and fulfillment, inventory-cost reduction, demand forecasting, operational performance management, continuous landed cost, and continuous inventory accounting and book-close readiness. These must share coherent demand, supply, allocation, and cost semantics so that local improvements do not create downstream failures.

Continuous landed cost replaces repeated manual classification and bill mapping with prepared, evidence-backed allocations. It learns customer-specific patterns from historical mappings and reviewer corrections, proposes how new charges relate to inventory bills, and executes approved allocations through the existing costing capabilities. Unmatched charges and incomplete cost allocation become visible operational and close-readiness exceptions.

Continuous inventory should help accountants close the books automatically after review. Throughout the period, the system should reconcile inventory activity with accounting, identify incomplete costing and posting, prepare corrections and supporting evidence, and maintain a current close package. After the accountant approves the defined package and scope, the system should execute authorized close steps, verify the results, and record completion. Inventory close contributes to the enterprise book close; full-period closure also depends on completion of other required accounting workstreams.

The first proof of the strategy should be a complete, approval-based material-shortage resolution experience. It should establish reliable facts, feasible plans, controlled execution, and verified outcomes before autonomy expands.

Strategy at a glance. The diagrams in this paper show business responsibilities and decision flows, not proposed service boundaries or deployment architecture.



1. Current state

This description reflects the platform briefing provided for this strategy. It is a business and capability baseline, not an independently verified inventory of implementation details.

1.1 Existing capabilities and responsibilities

Capability	Current scope
Item Master	Inventory items, non-inventory/service items, assembly items and associated BOMs, units of measure, product categories, preferred vendors, and alternate vendors.
IOM	Inventory locations, serial/lot definitions, item receipts, manufacturing orders, build assembly transactions, quantity and value adjustments, inventory transfers, pick/pack/ship, cycle counting, and fulfillment orders. Exact cross-service posting responsibilities require confirmation.
1TS	Order and financial transactions; synchronous inventory quantity computation on transaction save; FIFO and moving average cost (MAC).
Landed cost	Supported through a manual process: customers configure relevant categories/chart-of-accounts mappings and mark items or accounts for landed cost, such as shipping, freight, or insurance. When those charges are billed, users manually associate them with one or more inventory bills. The owning capability, available allocation bases, and treatment of estimated versus finalized cost require confirmation.
Period-close controls	Current support for close checklists, approval, period locking, reopening, and cross-workstream close coordination has not been specified. These are capabilities to confirm during discovery.
BI/reporting	Separate reporting capability.
Search and audit	Separate supporting capabilities.

1TS includes estimates, sales orders, invoices, sales receipts, refund receipts, purchase orders, bills, expenses, checks, journal entries, deposits, and prepayments.

Manufacturing orders are non-posting transactions. Completion leads to a build assembly transaction that posts to the books. Fulfillment orders are non-posting transactions that send sales orders for fulfillment.

Current capability landscape. Grouping below reflects the briefing; it does not imply that the full transaction lifecycle is owned by a single service.



1.2 Inventory dimensions and activities

Locations have two levels: **L1** represents a warehouse, shop floor, or distribution center; **L2** represents a bin within an L1. Each inventory transaction line maps to one L1. Its L2 and serial/lot quantity breakdown is represented one-to-many through inventory activities.

Serial or lot tracking can be enabled per item. An item currently supports one of these tracking modes, not both simultaneously.

Activity	Current meaning	Strategic significance
Inventory assignment	Soft allocation for a purpose such as sales-order fulfillment, transfer, or manufacturing.	Represents intended use that planning must respect and, where allowed, reconsider.
Inventory commitment	Hard reservation, such as stock sent for picking or issued for a project/manufacturing order.	Protects work already committed to execution.
Inventory movement	Operational inventory ledger activity, whose timeline can differ from the accounting ledger.	Establishes recorded physical changes and their relationship to transaction and financial state.

1.3 What this foundation enables

The platform already distinguishes intent, reservation, movement, and posting. That separation is valuable: preparing a manufacturing order, issuing material, recording output, and posting its value are different business events.

The next strategic capability is coordinated management across these events. The briefing does not establish whether forecasting, finite-capacity planning, optimization, or cross-service automation already exist. This paper treats them as capabilities to assess and develop, not confirmed gaps.

2. Vision and customer promise

Vision: An enterprise inventory and manufacturing system that continuously anticipates needs, prepares the best feasible response, carries authorized work through completion, and keeps inventory accounting ready for an accountant-reviewed, automatically executed book close.

The operating cycle is:

Observe / understand / project / prepare / authorize / execute / verify / improve.

“Continuous” means responding within the time required by the decision. A picking commitment may affect availability immediately. Forecasts and policy parameters may update at a slower cadence. Production schedules need stability windows, and significant changes need thresholds that prevent unnecessary replanning.

The customer promise has four parts:

1. **Prepared decisions:** The investigation, calculations, alternatives, and required changes are ready before approval is requested.
2. **Accountable follow-through:** The system coordinates digital transactions, physical tasks, and external dependencies after authorization.
3. **Visible control:** The customer can understand, modify, reject, delegate, or revoke authority for actions.
4. **Verified results:** Completion is supported by operational evidence and the intended business outcome.

A transfer is not complete merely because a transaction was created. Depending on the case, completion may require dispatch, receipt, eligible material at the destination, and reconciliation with posting state.

The corresponding accountant experience is: **“Your inventory close package is prepared. Review the proposed entries, reconciliations, and exceptions; approve it, and we will complete the authorized close work.”** Continuous preparation reduces period-end investigation while preserving an explicit review and close decision.

3. Strategic outcomes

The system should improve customer service and operational reliability while controlling inventory investment and execution cost.

Outcome	What should improve
Reliable availability	Fewer stockouts, failed picks, and discrepancies between expected and usable inventory.
Reliable production	More manufacturing orders material-ready on time; less shortage-related interruption.
Reliable fulfillment	Better order fill and shipment performance; earlier recovery from supply disruptions.
Efficient inventory investment	Less excess, aging, and avoidable buffer stock without degrading agreed service.
Efficient execution	Less manual investigation, coordination, expediting, and repetitive approval work.
Continuous close readiness	Fewer unresolved accounting exceptions, faster reviewed close, less manual reconciliation, and fewer post-close corrections.
Trustworthy automation	Authorized actions execute correctly, exceptions surface promptly, and outcomes remain explainable.
Business-process fit	Recommendations reflect the customer's item-specific and seasonal operating practices; corrections persist at the intended scope and reduce repeated mistakes.

Targets should be baselined by customer segment, site, and workflow before numerical commitments are made. Service, cost, and working-capital priorities should be explicit customer policies. There is no universal optimum across all enterprises.

4. Strategic pillar: trustworthy operational understanding

Continuous decisions depend on a shared understanding of inventory and its future uses. The platform needs consistent business definitions across operational views, planning, reporting, and automation.

4.1 Distinguish facts, observations, expectations, and proposals

A posted receipt, a physical count, a supplier's promised arrival date, a predicted delay, and a proposed transfer have different evidentiary status. Each must retain its source, relevant time, and reliability. Predictions and proposals must not silently become confirmed physical or financial facts.

Current inventory is not a single quantity. On-hand, eligible, assigned, committed, and projected available quantities answer different questions. Assignment-to-commitment and commitment-to-movement transitions must have explicit semantics to prevent demand from being counted twice.

4.2 Preserve operational and accounting time

The system must relate physical execution to transaction and accounting state without assuming they advance together. Backdated changes, delayed postings, reversals, and corrections may affect historical views and active decisions differently.

Disagreement becomes a reconciliation workflow with supporting evidence. Automatically inventing an adjustment to make two views agree is not an acceptable resolution.

4.3 Represent dependencies and uncertainty

The system needs to connect inventory and expected supply to the manufacturing and customer requirements they support. A component delay should reveal the affected production and fulfillment commitments.

Future supply should carry uncertainty. A purchase order is expected supply, not physically available stock. Planning must account for lead-time variability, receipt status, and evidence freshness.

5. Strategic pillar: coordinated demand and supply planning

Auto reorder, auto build, and fulfillment routing must share a consistent supply-demand view. Without coordination, they can create duplicate supply or compete for the same inventory.

5.1 Demand forecasting

Forecast independent demand at an item, location, and time granularity suited to the decision. Capture forecast versions, planning horizons, uncertainty, and human overrides.

Demand signals should distinguish sales demand from internal transfers and manufacturing component consumption. Fulfilled sales alone can understate demand during stockouts. New items, intermittent demand, returns, and unusual promotions require explicit treatment rather than one forecasting method for every item.

Actual orders should consume forecast demand under defined rules. Assembly demand should generate dependent component requirements through the BOM. Forecasts, sales orders, and component requirements must not be blindly added together.

5.2 Inventory policies

Support service targets, safety-stock policies, reorder parameters, lot sizes, purchasing constraints, preferred suppliers, and inventory budgets. The system should recommend policy changes using observed outcomes while preserving customer authority over objectives and material changes.

Policy adaptation needs stability. A short-lived anomaly should not repeatedly change safety stock or reorder behavior.

Policies must support mixed operating modes within one customer: bulk replenishment for some items, production or purchasing against individual sales orders for others, and seasonal changes to either. Demand forecasts inform these

policies; a forecast alone does not authorize buying or building ahead of orders.

5.3 Coordinated alternatives

For a requirement, evaluate feasible combinations of buy, build, transfer, reallocate, and reschedule. Compare service, timing, capacity, cost, and downstream impact. Protect hard commitments and make any proposed disruption visible.

Planning can be hierarchical: enterprise policies set objectives and limits; sites make operational decisions within them. Conflicts over shared stock, capacity, or budgets require explicit arbitration.

One coordinated plan across operating modes. The applicable mode can differ by item, site, and season. Forecast consumption and supply netting prevent duplicate requirements and duplicate supply.



6. Strategic pillar: customer-specific learning and steering

The system should learn the customer's business process rather than apply one universal replenishment or production pattern. Learning and customer steering must influence planning, routing, landed cost, and close preparation through a shared, inspectable understanding of the business.

6.1 Learn at the right scope

Scope	Examples of what the system should understand
Business or legal entity	Service priorities, buying cadence, cash constraints, approval practices, and default operating modes.
Item or item group	Bulk order quantities, production batches, make-to-stock versus make-to-order, purchase-to-order, and permitted substitutes.
Location or supplier	Site-specific buffers, supplier minimums, consolidation practices, and local lead times.
Season or business event	Advance production before a peak, seasonal batch changes, holiday shutdowns, and post-season stock reduction.
Order or customer context	Customer-specific fulfillment terms, dedicated procurement, and exceptions for urgent or customized orders.

The same item may be built in bulk before a peak season and produced against confirmed orders during the rest of the year. A customer can also stock common components while assembling finished goods only for sales orders. These are legitimate hybrid processes, not inconsistencies to normalize away.

For order-driven supply, preserve the relationship between the sales order and its purchase/build requirement. For pooled supply, coordinate quantities across multiple requirements. Aggregating orders into a batch must respect the customer's rules for timing, allocation, and traceability.

6.2 Combine declared process, observed practice, and outcomes

Use customer instructions, approved configuration, historical transactions, plan edits, rejected proposals, and verified results as different learning signals. Store learned patterns with scope, evidence, effective dates, confidence, and review history.

An observed pattern is a hypothesis about practice. It is not automatically a constraint or permission. A large purchase may reflect an unusual discount or emergency rather than a standing preference for bulk ordering. A seasonal pattern requires sufficient seasonal evidence or an explicit customer declaration.

Keep three concepts distinct: **what the customer wants, what the system has observed, and what the system is authorized to do.** Feasible plans must satisfy applicable controls and explicit authorized rules. Within those boundaries, current scoped instructions take precedence over inferred habits. If customer rules conflict, explain the conflict and obtain resolution from the appropriate owner instead of silently choosing one.

The agent may propose a better process, showing the expected tradeoffs, but should not silently replace a customer's declared operating model with an inferred optimum.

6.3 Let customers correct and teach the agent

Customers should be able to steer the agent in the context of a proposed action, through ordinary language or direct edits. For example:

> “For this item, build only against confirmed sales orders. From September through November, combine those orders into weekly batches of up to 500.”

The system should translate this into a visible business rule, show its interpretation, and apply it through the appropriate authorization path. Distinguish a one-time correction from an ongoing instruction. When scope is ambiguous, clarify whether it applies to this action, item, item group, location, season, or the whole business; do not silently generalize it.

Show the immediate effect on the current plan and identify other open proposals that now need reconsideration. Existing commitments and approved work require their normal change controls. A correction should not silently cancel work already underway.

Give customers a place to inspect, edit, suspend, or remove what the agent has learned and been told. Preserve why a rule exists and who may change it. Ask for clarification only where needed to avoid a material misinterpretation; reuse established context elsewhere.

6.4 Serve customers without history

For new customers with no usable platform history, start with a lightweight process conversation and representative examples. Establish which items are stocked, bought or built to order, ordered in bulk, or handled seasonally, together with key constraints and approval boundaries.

Offer editable starting policies with explicit assumptions. Where available, customer-provided operating procedures and imported history can accelerate setup, but should not be prerequisites. Demonstrate the agent's interpretation through sample plans before consequential execution, then improve it through reviewed live proposals.

Sparse data should produce visible uncertainty and targeted questions, not invented confidence. New items and new locations need the same treatment even within established customers.

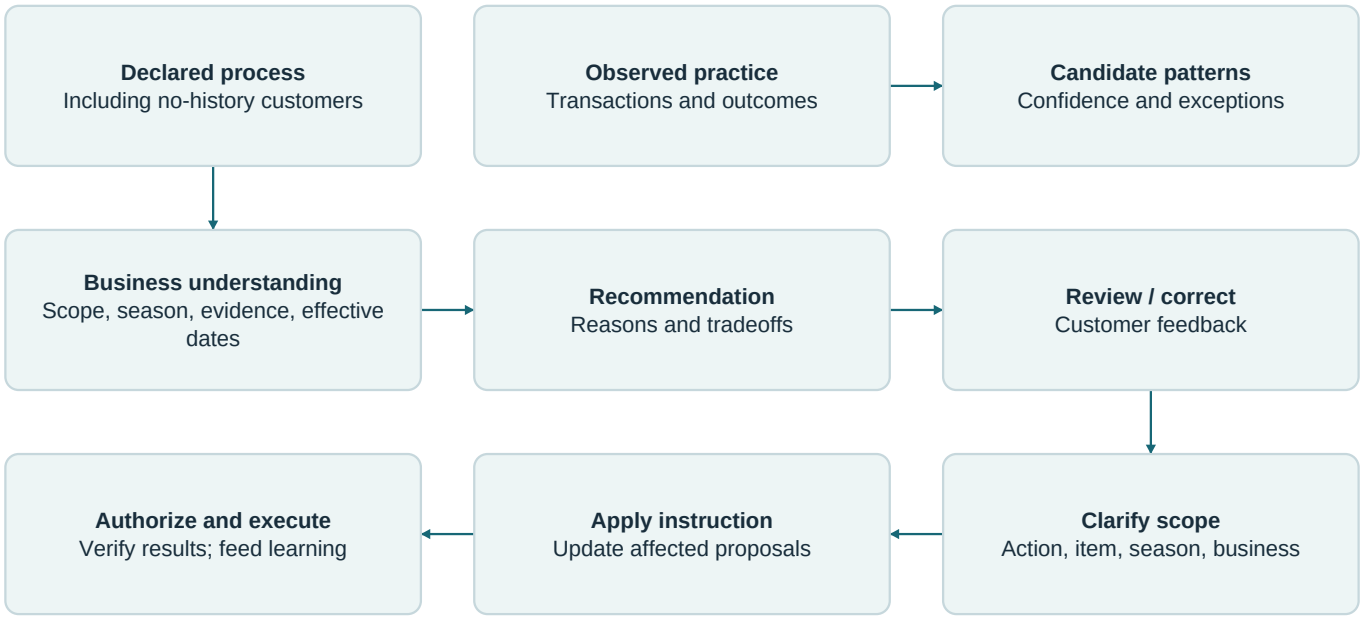
6.5 Adapt without repeatedly relearning mistakes

Detect changes in demand, lead times, customer corrections, and outcomes. Separate recurring seasonality from temporary exceptions and lasting process changes. Revisit stale learned patterns while retaining explicit instructions until an authorized change or their stated expiry.

Evaluate whether learning improves both business outcomes and process fit. Track repeated errors after correction, rule adherence, proposal edit rates, and performance by item and season. Learn from outcomes and overrides as well as approvals, so the system does not merely reinforce its own previous recommendations.

Learning remains customer-scoped and must respect access boundaries. Updating a preference or learned pattern does not expand execution authority.

How the customer teaches the system. Observations improve recommendations; authorized instructions determine the intended process, and permissions control execution.



7. Strategic pillar: done-for-you operating experiences

7.1 Auto reorder

Detect a future shortage, validate outstanding supply, select an eligible vendor, calculate quantity and timing, and prepare a purchase order. Evaluate lead-time variability, order minimums and multiples, estimated landed cost, and destination requirements where data exists.

After authorization, follow the order through supplier confirmation, delay management, and receipt. Creating a PO resolves the purchasing action; it does not by itself resolve the original material need.

7.2 Auto builds

Translate finished-goods requirements into proposed manufacturing orders and coordinated material readiness. Respect BOM definitions, component eligibility, batch constraints, and actual production constraints where modeled.

Separate authority to create an order, release production, commit material, and post completed output. The build assembly transaction must reflect confirmed production through existing posting rules.

If the platform currently models only material feasibility, communicate that boundary. Capacity-feasible promises require suitable routing, resource, calendar, and duration information.

7.3 SO routing and fulfillment

Evaluate eligible inventory across L1 locations, competing commitments, customer dates, permitted splits, warehouse readiness, and transport options where available. Prepare the fulfillment route, assignments, and fulfillment orders, then follow pick/pack/ship execution.

Routing must respect the one-L1-per-transaction-line model and supported fulfillment relationships. Once stock is hard committed, changing the route requires an authorized operational transition. Planning cannot simply overwrite work already in progress.

7.4 Inventory-cost reduction

Identify excess, aging stock, duplicated buffers, avoidable expediting, and opportunities to use existing inventory. Prepare transfers, replenishment changes, purchase/build rescheduling, and disposition proposals.

Use landed cost consistently. Distinguish estimated acquisition cost, actual inventory valuation, and incremental cost of an action. Avoid double-counting freight or duties already included in landed cost. Financial valuation remains under the existing costing and posting capabilities.

Cost reduction must respect service and operational constraints. Report the difference between projected benefit, observed savings, and working-capital release.

7.5 Inventory accuracy and manufacturing readiness

Continuously detect discrepancies and readiness risks. Prepare targeted cycle counts, evidence-backed reconciliation, material transfers, and staging work. These experiences establish confidence in the data used by higher-autonomy purchasing and production decisions.

7.6 Continuous landed cost

The target experience is: **“We identified this freight charge, matched it to the inventory bills it supports, and prepared the allocation using your approved method. Review the evidence and cost impact; approve it, and we will apply and verify it.”**

Today, customers perform both setup and transaction-level matching manually. The continuous experience should address both while keeping them distinct:

- **Classification and configuration:** Understand which categories, chart-of-accounts entries, and charge items the customer uses for landed cost; suggest additions or corrections to that configuration.

- **Transaction matching and allocation:** Determine which specific inventory bills a charge relates to, the eligible amount, and how it should be distributed under the customer's approved method.

Marking an account or item as eligible does not establish that every charge on it belongs to a particular inventory bill. Matching a charge to bills also does not determine the correct allocation basis by itself.

Learn customer practice with explicit evidence

Learn from customer-scoped historical mappings, accepted proposals, reviewer edits, rejected matches, and subsequent corrections. Relevant patterns may include charge vendor, descriptions, account/item/category, inventory supplier, shipment or PO references, dates, destination, and recurring combinations of inventory bills, where those signals are available.

Use this history to suggest a customer's recurring practice and explain the supporting examples. Historical mappings may contain errors or one-off exceptions; repetition is evidence, not authority. Explicit approved rules take precedence over inferred patterns. Changed practices should be versioned, and contradictory evidence should reduce confidence or trigger review.

Keep confidence in classification, bill matching, and allocation method separate. A familiar freight vendor can support classification while leaving the affected shipment uncertain. A new customer should be able to start with explicit setup and reviewed proposals before sufficient history exists.

Prepare and complete the allocation

Step	Done-for-you responsibility
Detect	On new or changed bills, identify configured landed-cost charges and propose candidates that may have been missed in setup.
Match	Identify eligible inventory bills using document references, transaction relationships, and supporting customer patterns; show ambiguity and missing evidence.
Allocate	Prepare amounts across one or more inventory bills and their eligible lines where supported, using an approved allocation basis and available data.
Review	Show source charges, target bills, supporting evidence, allocation method, amounts, unallocated remainder, and expected cost effects.
Execute	Apply the approved mapping and allocation through existing owning services, preserving links to the source charge and authorization.
Verify	Confirm allocation totals, costing completion, resulting valuation/posting status, and any remaining exception.

Support partial charges and multiple charge bills against the same inventory bill without allocating a source amount twice. The allocated amount plus the explicitly unallocated remainder must reconcile to the eligible source amount, with currency and rounding treatment defined by the owning accounting capability.

Changes, credits, cancellations, and late charges should reopen the relevant allocation case for reassessment. Where stock has already been consumed or sold, surface the resulting cost impact through the existing costing rules rather than assuming all added cost remains in on-hand inventory. Closed-period effects follow the authorized correction or reopening process.

Graduate autonomy by mapping pattern

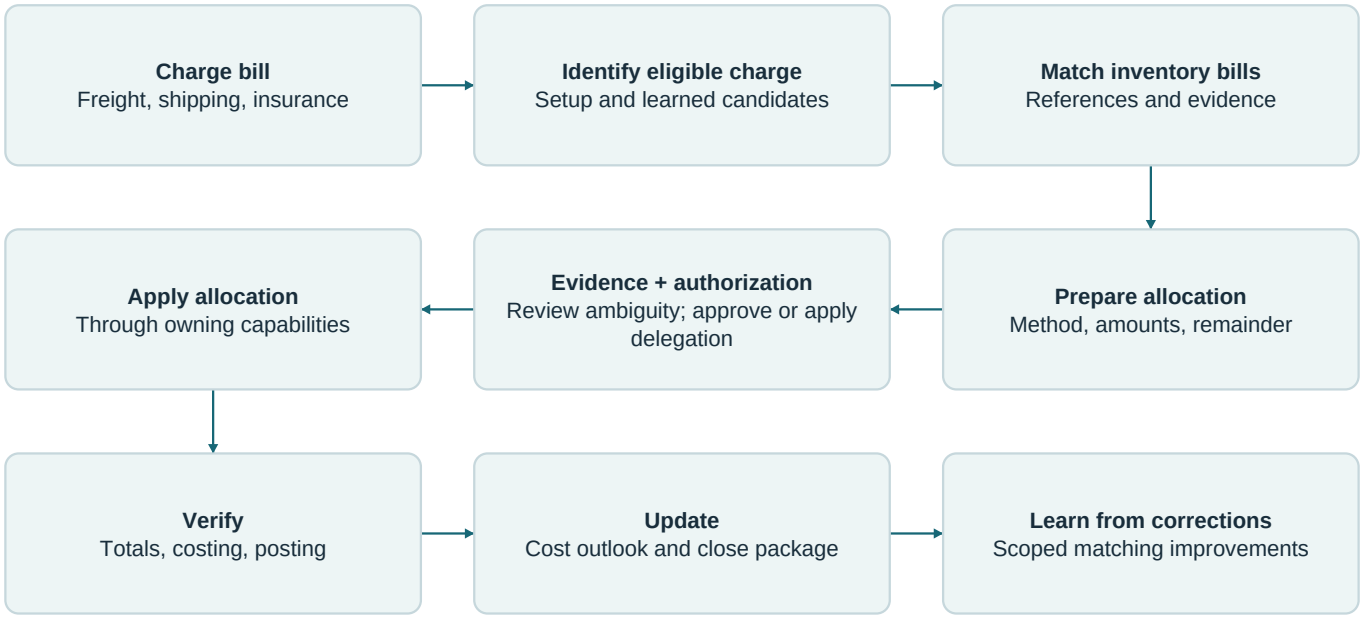
Begin with reviewed proposals. Customers may later delegate well-established combinations of charge type, vendor, reference quality, allocation basis, entity, and value limits. High model confidence alone does not grant authority. Configuration changes and transaction allocations should have separately scoped permissions.

Ambiguous matches, new cost types, missing allocation inputs, unusual amounts, or conflicts with approved rules should return to review. Routine cases can be completed automatically only within an explicit delegation, with traceable evidence and a supported correction path.

Connect landed cost to planning and close

Feed actual allocated costs back into future sourcing and replenishment estimates, while keeping estimated and finalized cost distinct. For accountants, expose unallocated charges, aging unmatched bills, pending costing, and changes awaiting approval as part of the close package. An apparently complete bill should not conceal incomplete landed-cost work.

From a new charge to verified landed cost. Configuration eligibility, bill matching, and allocation are separate decisions. Ambiguity returns to review.



7.7 Continuous inventory accounting and book close

Continuous inventory should continuously prepare the inventory-related accounting work needed to close the books. The objective is to resolve issues near their source, keep evidence current, and let accountants review a complete package before the system executes the approved close steps.

Continuous close readiness and formal period close are distinct. Readiness evolves throughout the period. Formal close is an authorized action for a defined entity, ledger, period, and scope. Inventory readiness alone cannot establish that the entire enterprise ledger is ready to close.

Work performed throughout the period

Workstream	Continuous system responsibility
Operational-to-financial reconciliation	Connect inventory activities and movements to relevant transactions and postings; identify missing, duplicate, delayed, or inconsistent records.
Quantity and valuation reconciliation	Reconcile inventory detail and valuation with applicable inventory control accounts, explaining differences rather than inserting balancing adjustments.
Cost completion	Identify pending FIFO/MAC processing, missing cost inputs, unmatched or partially allocated landed-cost charges, provisional costs, and changes whose financial effects remain unresolved.
Manufacturing accounting readiness	Track the relationship between component consumption, production completion, build assembly posting, and WIP or production-cost treatment where supported.
Cutoff and outstanding supply	Surface receipts, shipments, transfers in transit, returns, and supplier documents requiring period attribution or review under customer-approved accounting policies.
Adjustments and estimates	Prepare evidence-backed quantity/value corrections and proposed accruals or valuation adjustments where applicable, with the accounting policy and reviewer identified.
Exception resolution	Assign owners, deadlines, evidence requirements, and dependencies; follow unresolved cases through disposition.

These are proposed product responsibilities, not assertions that every accounting treatment or close capability exists today. Recognition, valuation, cutoff, and materiality rules must follow the customer's applicable accounting framework and approved policies. AI may assemble evidence and propose treatment; it does not establish accounting policy.

The accountant review package

Prepare a versioned package showing opening balances, movements and financial effects, closing balances, reconciliations, proposed entries, costing status, and unresolved exceptions. Each material balance and proposed adjustment should link to supporting transactions, inventory activities, calculations, and documents.

Distinguish blockers from policy-permitted estimates or exceptions. The package should explain the consequences of unresolved items and identify which require explicit accountant acceptance. Missing evidence must remain visible.

Approval binds to the package version, scope, proposed entries, and permitted tolerances. A relevant change before execution requires revalidation and, where outside those tolerances, renewed review. Operational approval of a receipt or build is not automatically approval of an accounting adjustment or period close.

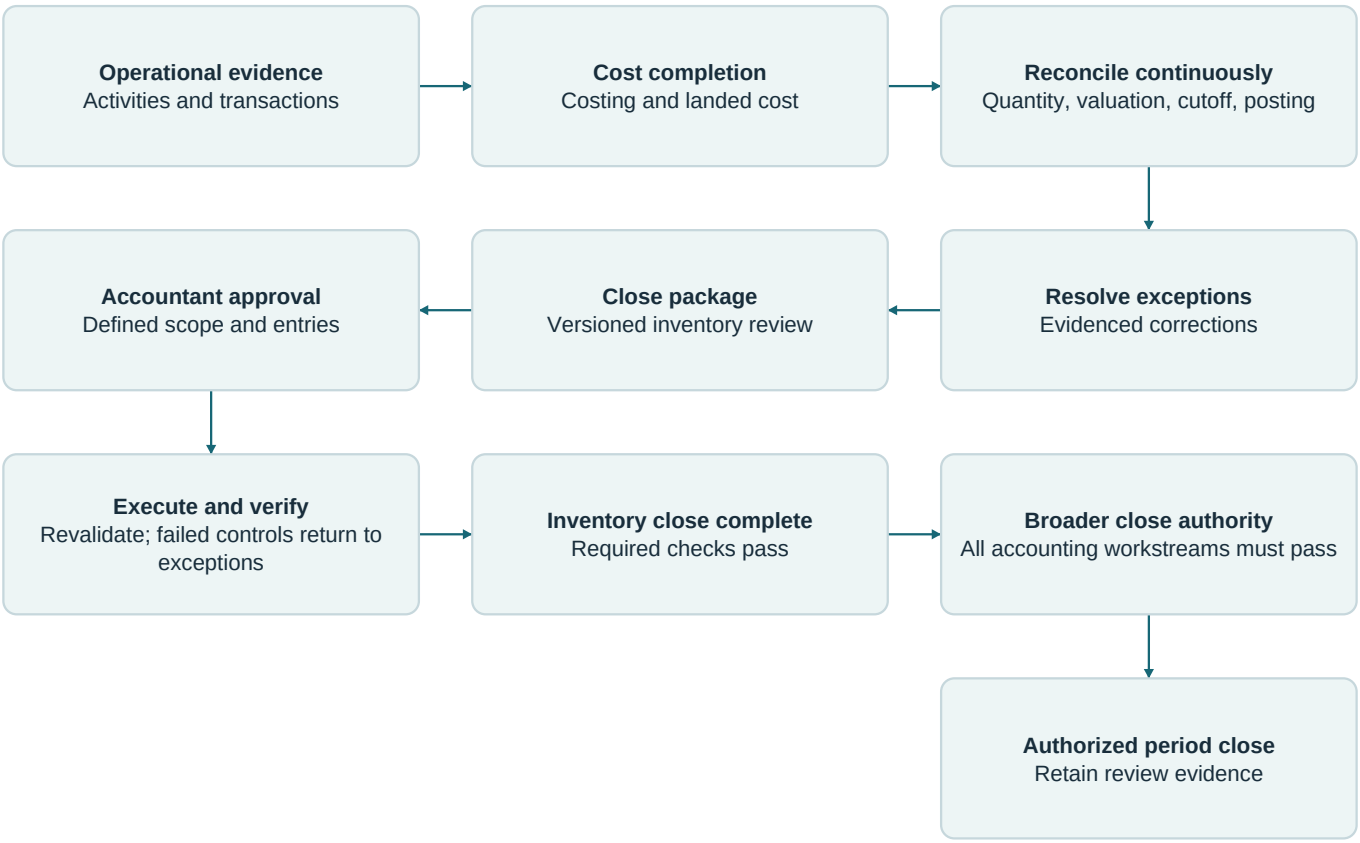
Automatic completion after review

After approval, the system should execute authorized entries through 1TS and other owning capabilities, confirm costing and posting completion, rerun required reconciliations, and complete the approved inventory-close tasks. Period locking or full book-close completion should occur only through the designated close authority after all required workstreams and controls have passed.

Partial success must remain explicit. A posted adjustment followed by a failed reconciliation leaves the close incomplete; recovery must not duplicate the entry or report success prematurely. Late or backdated changes affecting a closed period should trigger the applicable correction or authorized reopening process, preserving the original close evidence.

Where broader close coordination is provided elsewhere, continuous inventory should supply its reviewed package, dependency status, and verified completion to that capability. This lets inventory businesses progress toward automatic book close after review without treating inventory as the only accounting workstream.

Continuous preparation, reviewed close. Inventory close supplies one verified part of the broader book close.



8. Strategic pillar: approval first, graduated autonomy

The unit of approval is a complete, versioned plan tied to a business outcome. It should show the proposed changes, supporting evidence, expected impact, costs, uncertainties, dependencies, and tolerances.

Delegation level	Customer control	System responsibility
Approve each action	Approves a specific prepared action.	Executes it and verifies its result.
Approve a coordinated plan	Approves related actions and explicit tolerances.	Coordinates the approved plan and surfaces material deviations.
Delegate bounded actions	Defines permitted action types, scope, limits, and exclusions.	Executes qualifying plans and seeks approval for exceptions.
Delegate an objective	Defines an objective, allowed means, budgets, and escalation rules.	Coordinates permitted actions toward that objective and reports results.

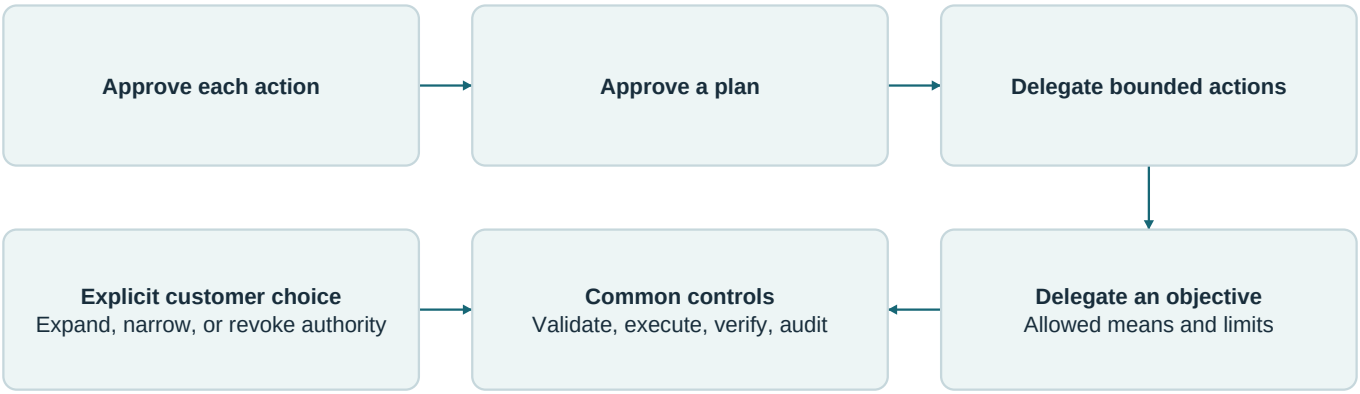
Delegation is specific to action category and operating context. Trust in internal bin replenishment does not authorize supplier purchases or financial adjustments.

For close workflows, preparation, reconciliation, adjustment posting, and formal close are separately scoped authorities. The initial experience retains accountant review of the close package; routine preparatory actions can gain bounded autonomy independently. Final close authority is never inferred from repeated operational approvals.

Trust should be assessed through recommendation quality, execution reliability, verified outcomes, and appropriate escalation under uncertainty. Approval rates alone are insufficient. The system may recommend expanded authority; the customer explicitly grants it.

Approval should remain valid only within its scope and tolerances. Material changes require reconsideration. Revocation must stop future unauthorized steps and identify already committed work requiring intervention. Cumulative limits should prevent many small actions from exceeding an intended budget.

The same execution discipline at every autonomy level. Progression is explicit and specific to an action category; customers can narrow or revoke delegation.



9. Strategic pillar: AI with enforceable business correctness

AI should interpret unstructured information, investigate cases, develop alternatives, explain tradeoffs, and coordinate work. Quantity arithmetic, eligibility, reservation integrity, permissions, costing, and posting rules require explicit enforceable logic.

The strategic separation of responsibilities is:

- **AI and predictive models:** interpretation, estimation, candidate plans, and explanations.
- **Planning and domain validation:** feasibility, resource conflicts, constraints, and expected effects.
- **Existing authoritative services:** valid business transactions and state changes.
- **Execution coordination:** long-running work, recovery, exceptions, and follow-through.
- **Verification:** evidence that the intended result occurred.

External messages and documents can provide evidence; they cannot grant authority. Model outputs should be inspectable and evaluated before they affect operations. Sensitive data access must respect customer, legal-entity, site, and role boundaries.

The platform must remain useful when AI is unavailable. Existing operational transactions should continue, and authorized deterministic workflows should have defined continuation or pause behavior.

10. Strategic pillar: metrics as decision inputs and proof of value

Metrics should be available in operational experiences and historical BI with shared definitions. Every metric needs a defined grain, time basis, source, freshness, and drill-through. Forecast values must be distinguishable from actual outcomes.

Area	Core measures	Primary use
Inventory position	On-hand, eligible, assigned, committed, inbound, projected available	Allocation and readiness

Area	Core measures	Primary use
Inventory investment	Value, average inventory, turns, days of inventory, excess and aging value	Policy and cost optimization
Customer service	Fill rate, stockouts, backorders, on-time shipment; OTIF when delivery evidence exists	Service protection
Replenishment	Lead-time distribution, on-time receipts, overdue supply, expedites	Reorder and vendor decisions
Landed cost	Unallocated charge amount and age, eligible charge coverage, mapping acceptance/edit and correction rates, time to verified allocation, estimated/actual variance, unit landed cost, cost components	Mapping quality, sourcing economics, and close readiness
Manufacturing	Material-ready rate, shortage exposure, schedule adherence, throughput, WIP age, yield and scrap where captured	Build planning and recovery
Warehouse	Pick accuracy, fulfillment cycle time, backlog, dock-to-stock time, count variance	Routing and execution improvement
Forecasting	Bias, horizon-specific accuracy, uncertainty coverage	Forecast and buffer improvement
Learning and process fit	Repeated errors after correction, explicit-rule adherence, proposal edit rates, time to first useful proposal for new customers, outcomes by item and season	Personalization quality and customer steering
Automation	Approval edits, execution reliability, resolution time, verified benefit, policy exceptions	Autonomy and quality reviews
Inventory close	Reconciled balance coverage, unmatched amount and age, pending costing/postings, unresolved landed cost, close blockers, review effort, time to close, post-close adjustments and reopenings	Accountant review, close readiness, and reliability

Measurement must account for data coverage. A shipment timestamp cannot establish delivery performance, and sampled cycle-count accuracy should disclose its sample. Forecast evaluation should use historical forecast versions rather than revised forecasts that incorporate later information.

Customers should see current position, forward risks, prepared actions, work in progress, and verified results. Metrics should link to the cases and transactions that explain them.

Accountants should additionally see a close-readiness view by entity and period: completed reconciliations, unresolved amounts, missing evidence, proposed entries, approvals, and remaining dependencies. Any readiness percentage must disclose its denominator; it cannot override a blocking control or substitute for review.

11. Enterprise operating model

The system should make accountability explicit across purchasing, planning, warehouse operations, manufacturing, and finance. A coordinated plan may require several decision rights rather than a single universal approver.

Exceptions should have an owner, deadline, business impact, and next action. Approval requests should arrive with sufficient investigation to be actionable. Related decisions should be grouped when that improves judgment, while financial or operationally sensitive steps retain their required controls.

Customers need visibility into what was proposed, what was authorized, what changed, and what outcome occurred. Tenant isolation, access control, segregation of duties, auditability, and data-retention obligations apply to AI-assisted workflows as they do to existing transactions.

12. Delivery strategy and progression gates

Progress should be evidence-based rather than driven by feature count or autonomy level.

Stage	Scope	Exit evidence
Establish trustworthy semantics	Quantity/activity lifecycle, supply-demand reconciliation, baseline metrics, data-quality visibility	Representative scenarios reconcile without duplicate demand or supply.
Establish customer process fit	Declared and learned operating patterns, corrections with explicit scope, onboarding without history	Bulk, order-driven, hybrid, and seasonal scenarios behave as intended; corrections persist without changing authority.
Prove a complete experience	Approval-based manufacturing shortage resolution using available transfer/replenishment actions	Plans are feasible; execution survives exceptions; readiness is verified.
Prove continuous landed cost	Learn existing customer mappings, prepare reviewed allocations, execute and verify through existing costing capabilities	Amounts reconcile, duplicate allocation is prevented, corrections are handled, and unmatched charges remain visible.
Prove reviewed inventory close	Continuous reconciliation, versioned accountant review package, and automatic execution of approved close steps	Approved entries post once, reconciliations pass, exceptions remain visible, and inventory completion is distinguished from full book close.
Expand prepared operations	Auto reorder, auto-build planning, SO routing and fulfillment	Shared-resource conflicts are managed; outcomes improve against baseline.
Improve planning economics	Forecast-driven policies and coordinated cost optimization	Service targets hold while measured inventory or execution economics improve.
Expand bounded delegation	Proven action categories within explicit limits	Sustained reliability, effective revocation, and appropriate exception escalation.

These are progression gates, not a claim that forecasting or measurement waits until a later phase. Both should be developed early enough to support the first chosen experience. Bounded delegation may become appropriate for one mature workflow while others remain approval-based.

Close readiness should be developed alongside operational workflows because the same receipts, movements, builds, costing changes, and corrections drive both. Inventory close is a first-class outcome, not a reporting feature added after operational automation.

Evaluation should include historical replay, shadow proposals, customer-reviewed pilots, and measured live outcomes. Test conditions should include stale data, concurrent claims on stock, partial receipts, changed approvals, delayed posting, reversals, and unavailable external systems.

13. Main risks and strategic responses

Risk	Response
Inaccurate records produce confident bad plans	Expose data quality, reconcile discrepancies, and require stronger evidence for consequential actions.
Buying, building, and routing optimize independently	Share demand/supply semantics, commitments, budgets, and conflict resolution.
Frequent replanning destabilizes operations	Use commitment windows, tolerances, and materiality thresholds.
Approval becomes a new administrative burden	Prepare complete decisions, group related work, and delegate proven routine actions.
Automation claims completion too early	Define business completion criteria and verify physical and financial milestones as required.
Cost reduction degrades service	Optimize within explicit service constraints and measure both outcomes.
AI capability exceeds operational controls	Keep authority and business invariants enforceable independently of model behavior.
Product promises exceed available data	Gate capabilities by evidence coverage and disclose feasibility boundaries.
Inventory readiness is mistaken for full book-close readiness	Make close scope explicit and require all applicable accounting workstreams before formal period closure.
Late changes invalidate reviewed balances	Version the close package, revalidate before execution, and route closed-period changes through authorized correction or reopening.
Learned landed-cost mappings repeat historical errors	Prefer approved rules, preserve evidence, learn from corrections, and evaluate classification, matching, and allocation separately.
One-off behavior becomes a standing business rule	Distinguish observed patterns, explicit instructions, and temporary exceptions; retain scope, evidence, and effective dates.
Learning reinforces the agent's own mistakes	Evaluate overrides, rejected proposals, and verified outcomes; test adherence after corrections across items and seasons.

14. Strategy review decisions

The following decisions should be resolved during review, before implementation architecture is specified:

1. **Initial customer and operating profile:** Which enterprise segment, manufacturing mode, and warehouse complexity should guide the first release?
2. **First flagship experience:** Confirm manufacturing material readiness and shortage resolution, or select a different first outcome based on customer value and data readiness.
3. **Objective hierarchy:** How should service, production continuity, inventory investment, and execution cost be prioritized when they conflict?

4. **Meaning of auto build:** Confirm the intended scope across order creation, release, material execution, and completed-output posting.
5. **Initial autonomy boundary:** Which actions require individual approval, and which existing customer policies may already authorize bounded execution?
6. **Planning scope:** Confirm whether finite capacity, transport constraints, supplier confirmation, and delivery evidence are available or future capabilities.
7. **Proof of value:** Agree baseline measures, evaluation cohorts, and release criteria without promising unsupported numerical gains.
8. **Close scope and authority:** Confirm the inventory-close scope, broader book-close owner, accountant review roles, accounting policies, materiality thresholds, and period lock/reopen capabilities.
9. **Landed-cost learning and allocation scope:** Confirm available mapping history, document references, supported allocation bases and granularity, correction behavior, and the limits for delegated classification and allocation.
10. **Customer steering and learning:** Confirm who may set business-process rules, how overlapping instructions are resolved, and the minimum setup experience for customers without history.

15. Boundary with implementation architecture

Approval of this paper should establish the vision, operating model, capability scope, correctness principles, and delivery priorities. It should not commit to a cloud vendor, database, event platform, workflow engine, model provider, service decomposition, or deployment topology.

The next architecture phase should map these decisions to authoritative ownership, transaction and activity lifecycle contracts, integrations, consistency boundaries, plan and authorization schemas, execution recovery, metric lineage, security, observability, and migration. Particular attention is required for the boundary between IOM activities and 1TS quantity computation, costing, and posting, together with close-package versioning and the authority for period closure and reopening.

The strategic commitment is a system that prepares and completes coordinated operational and inventory-accounting work on the customer's behalf, keeps the books continuously ready for review, and executes approved close steps with trustworthy evidence, visible control, and measurable outcomes.